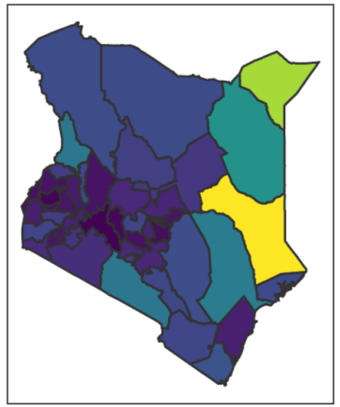
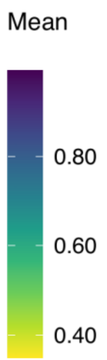
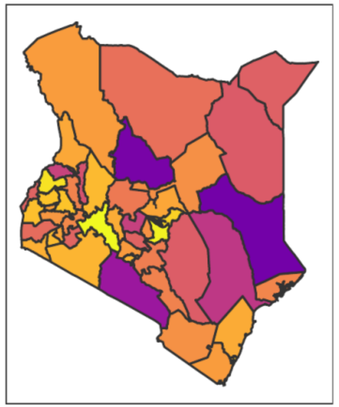
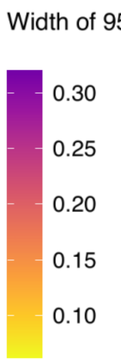
Direct, admin 1

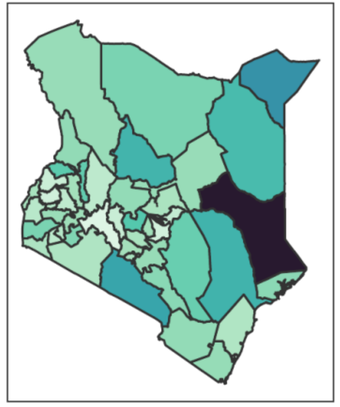
Without covariates

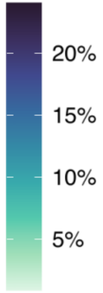


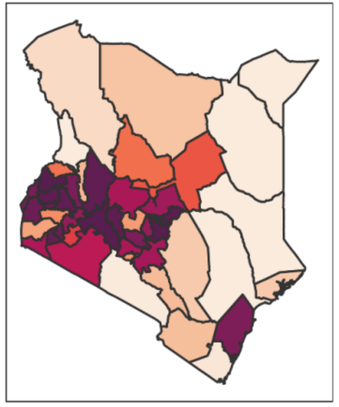


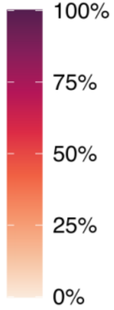




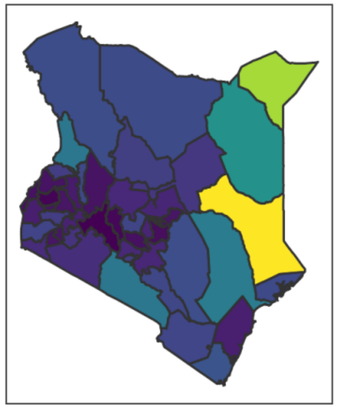


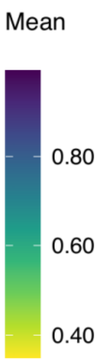


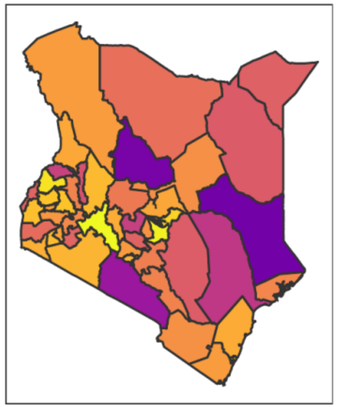


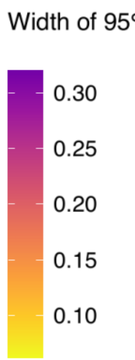


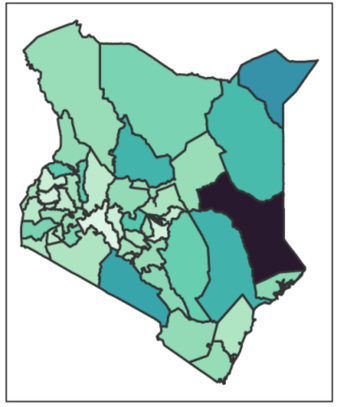
With covariates

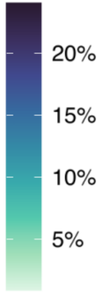


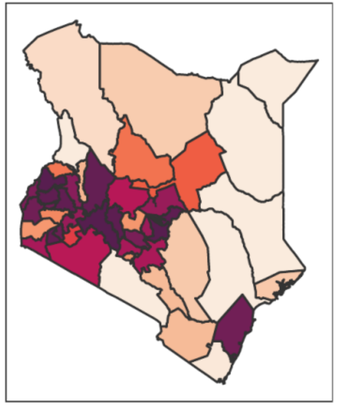


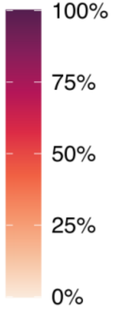






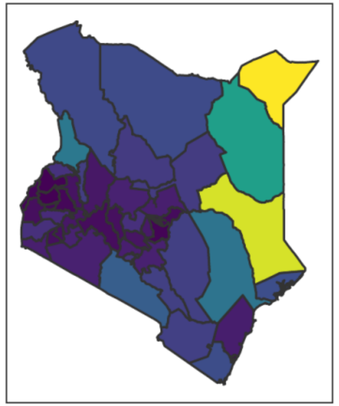


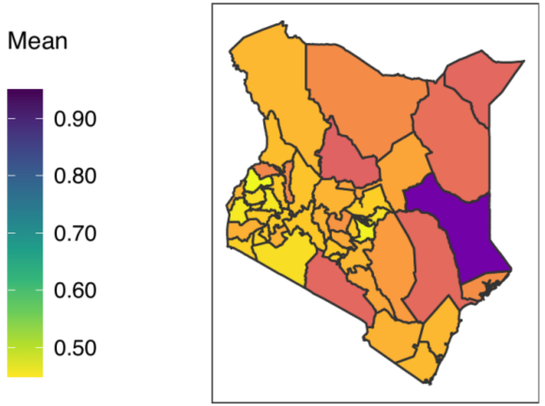


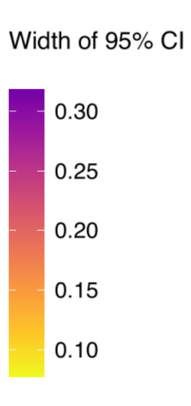


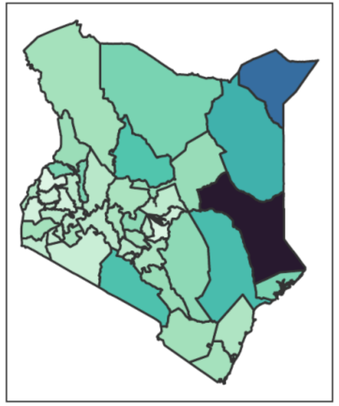
Area-level (Fay-Herriot), admin 1

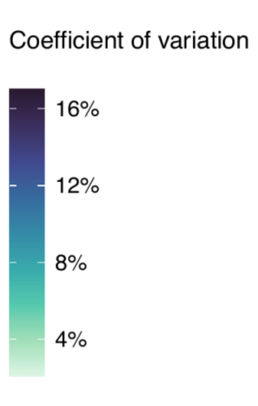
Without covariates

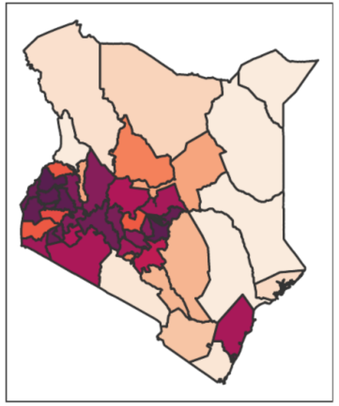


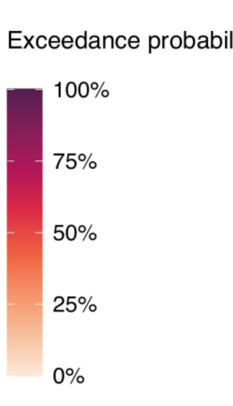




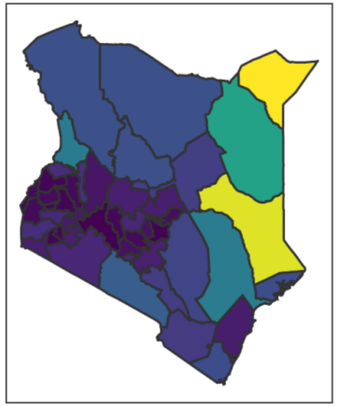


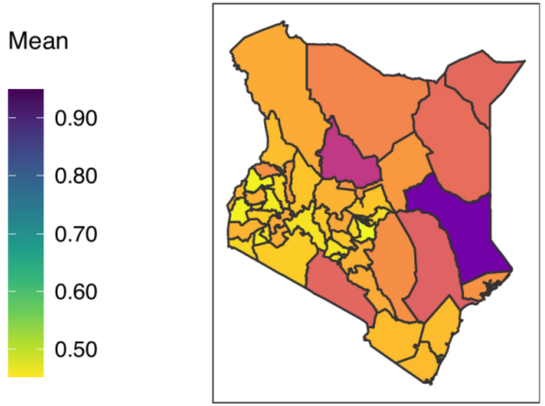


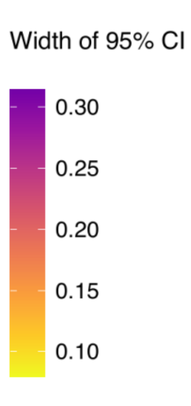


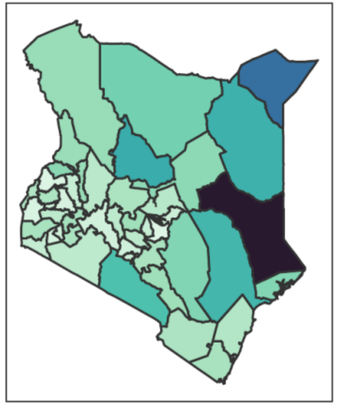


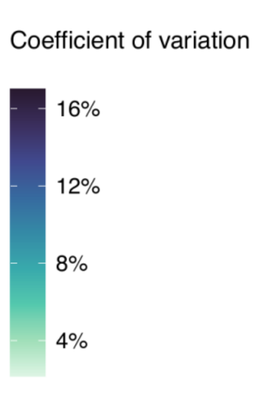
With covariates

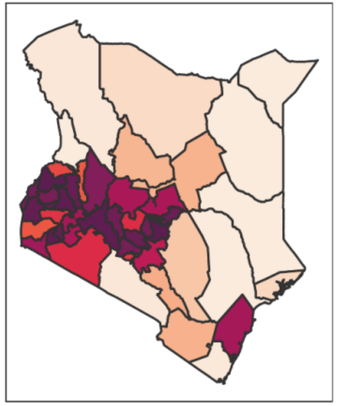


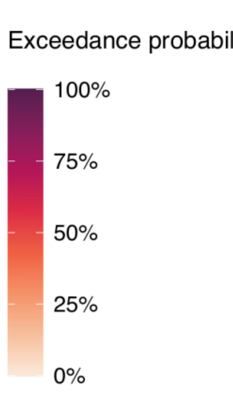






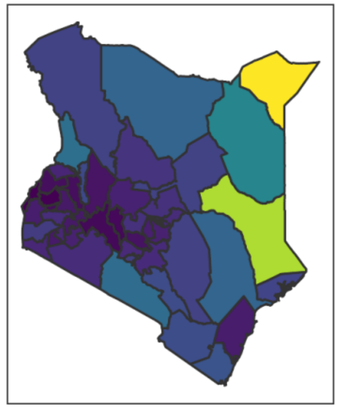


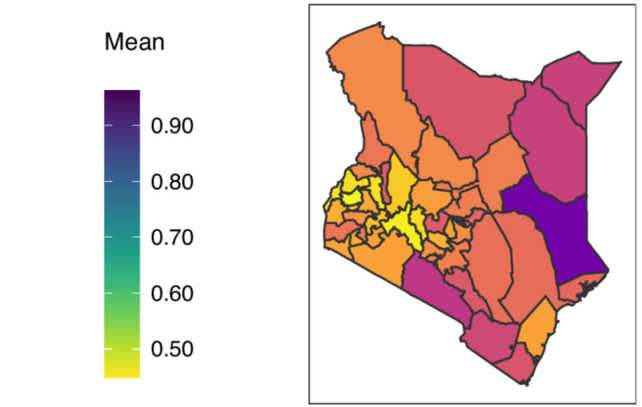


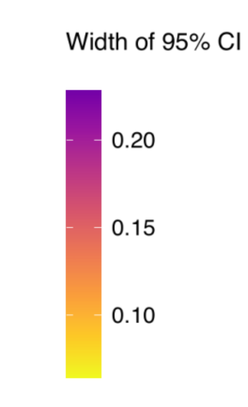


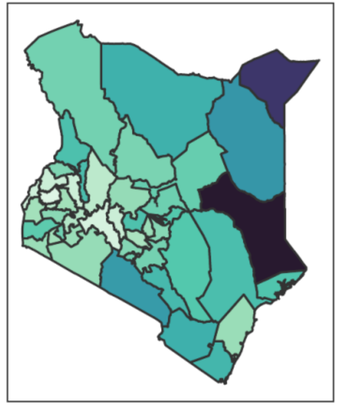
Unit-level, admin 1

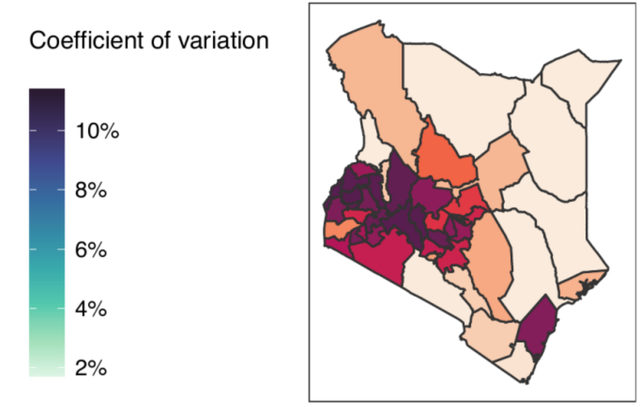
Without covariates

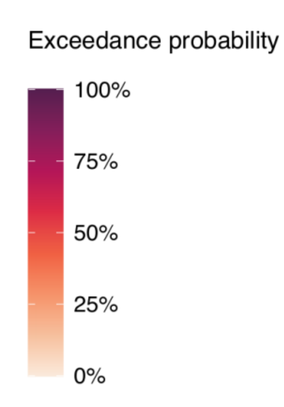




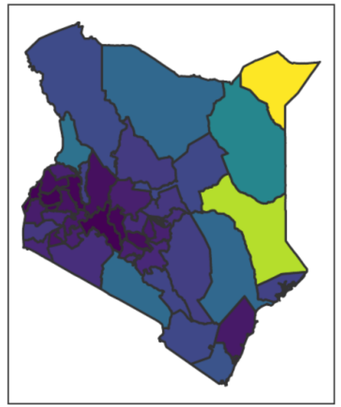


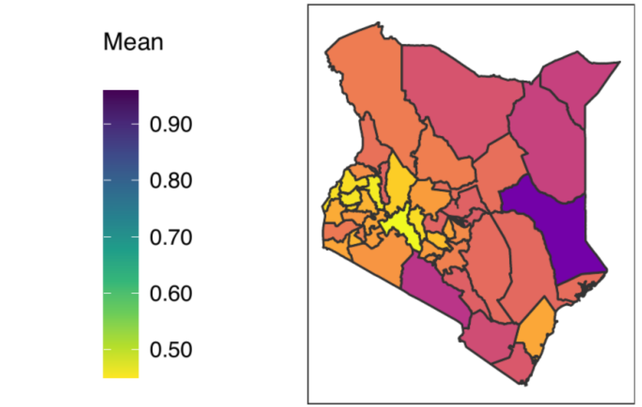


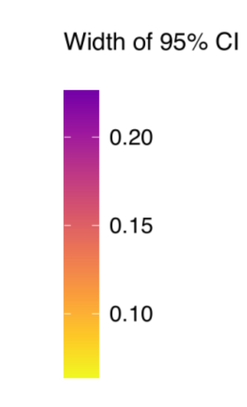


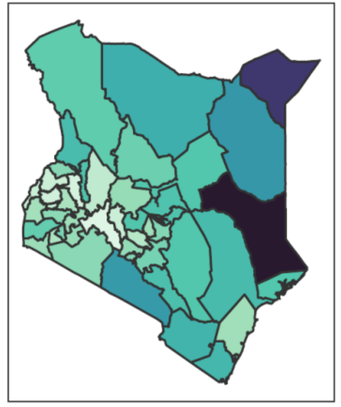


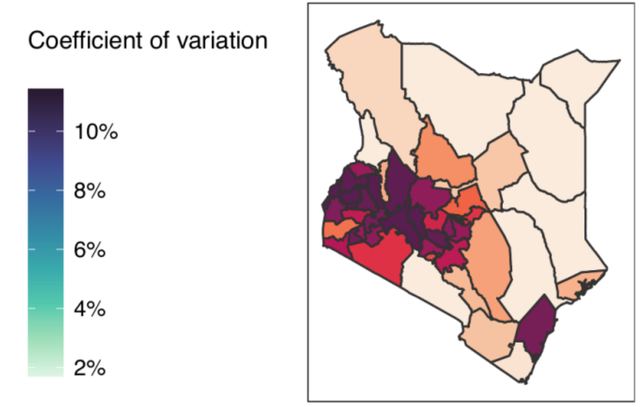
With covariates

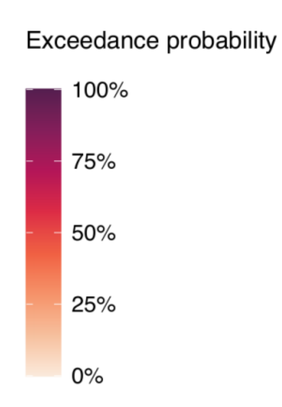






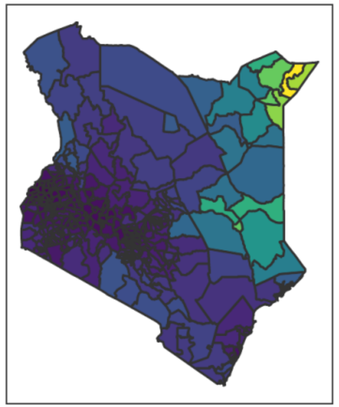


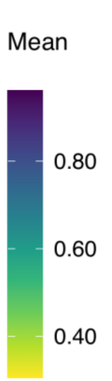


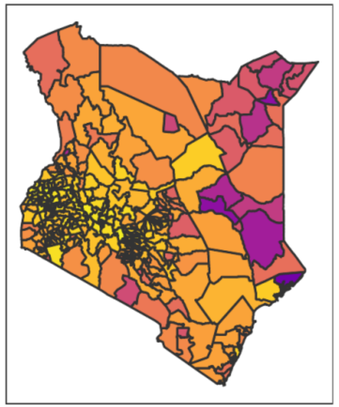


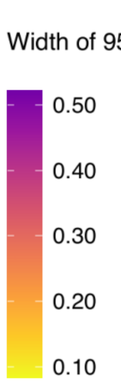
Unit-level, admin 2

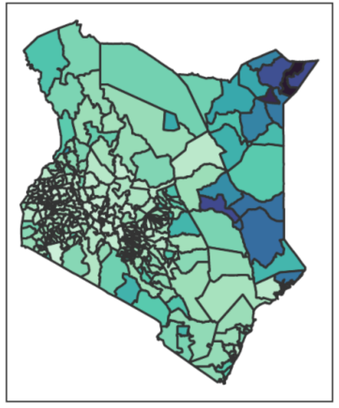
Without covariates

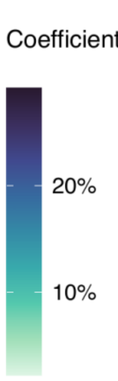


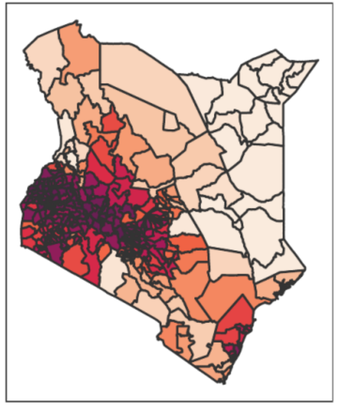


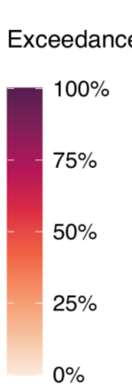




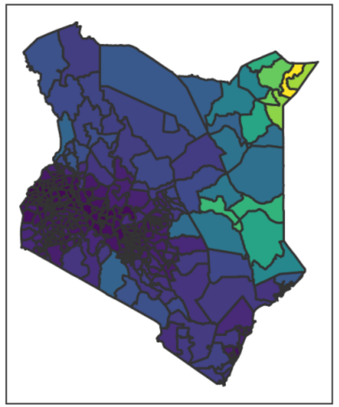


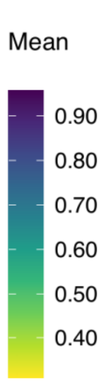


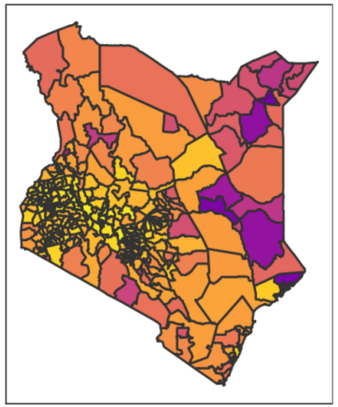


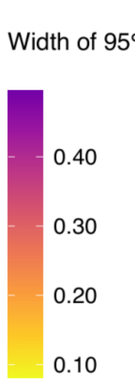


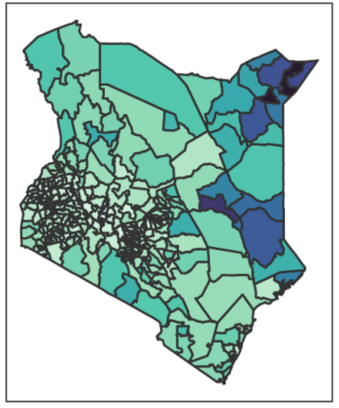
With covariates

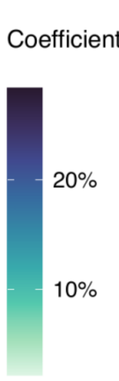


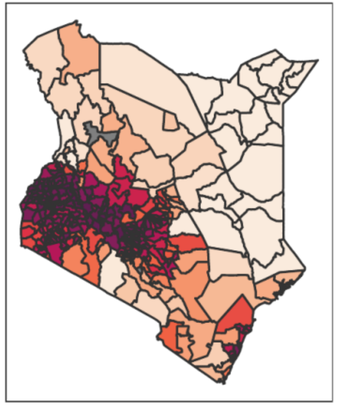


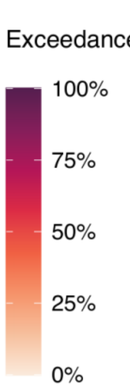


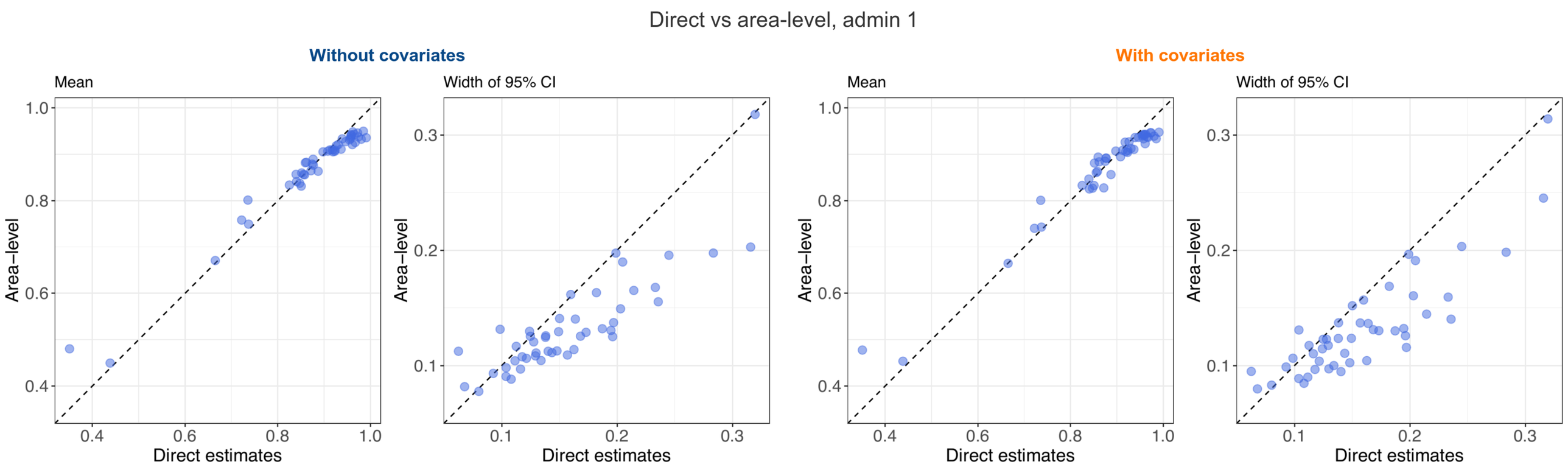


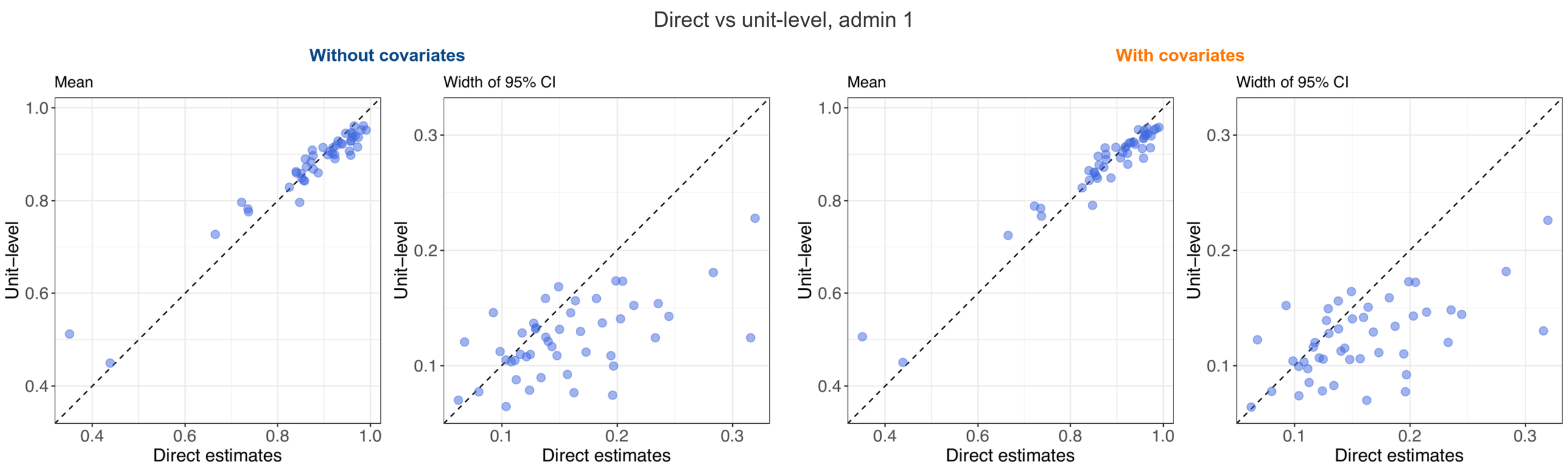









Direct vs area-level, admin 1
Without covariates
With covariates
Mean
Width of 95% CI
Mean
Width of 95% CI
Area−level
Direct estimates
1.0
0.8
0.6
0.4
0.3
0.2
0.1


Direct vs unit-level, admin 1
Without covariates
With covariates
Mean
Width of 95% CI
Mean
Width of 95% CI
Unit−level
Direct estimates
1.0
0.8
0.6
0.4
0.3
0.2
0.1